SIMATS ENGINEERING
ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	CSA1202	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL2-COMPARATIVE ANALYSIS
Weightage:	30%	Total Marks:	25
CO4 Statement:	*Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.*		
Student Name:	S.GAYATHERI	Reg.No.:	192521306
Department:	B.Tech Information Technology	Date:	22-08-2026

ANNEXURE A
COMPARATIVE ANALYSIS

1.Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.

Cache memory is a small, high-speed memory located close to the CPU. It stores frequently accessed instructions and data so that the processor does not have to access slower main memory every time.

Comparison

Feature	L1 Cache	L2 Cache	L3 Cache
Location	Inside each CPU core	Inside/close to each co	Usually shared among
Size	Smallest	Larger than L1	Largest
Latency	Lowest	Higher than L1	Higher than L2
Bandwidth	Highest	High	Lower than L1/L2
Throughput	Highest	High	Moderate
Access frequency	Very frequent	Frequent	Less frequent
Cost per bit	Highest	High	Lower
Main purpose	Immediate data/instructions	Backup for L1	Backup for L1 and L2

L1 Cache

L1 is the **fastest and smallest cache**. It is usually divided into:

- **L1 Instruction Cache** – stores instructions.
- **L1 Data Cache** – stores frequently used data.

Because L1 is very close to the processor, its latency is extremely low. It can supply data to the CPU at a very high bandwidth.

L2 Cache
L2 is larger than L1 but slightly slower. If the required data is not found in L1, the processor checks L2. L2 provides a balance between **capacity and speed**. It reduces the number of accesses to the slower L3 cache and DRAM.

L3 Cache
L3 is generally much larger and is often shared by multiple processor cores. It has higher latency than L1 and L2 but can store much more data.
For example:
CPU → L1 → L2 → L3 → DRAM

If data is found in L1, the CPU gets it very quickly. If it is not found there, it moves down the hierarchy.
Impact on Processor Performance
A cache **hit** means the required data is found in the cache. A **cache miss** means the processor must search the next memory level.

A high cache-hit rate results in:

- Lower average memory access time
- Higher CPU utilization
- Higher instruction execution rate
- Better system throughput

Therefore, **L1 gives the greatest speed advantage**, L2 provides additional capacity with slightly higher latency, and L3 reduces expensive DRAM accesses.

Conclusion
The three cache levels form a hierarchy that balances **speed, capacity, and cost**. Increasing cache size can improve performance, but larger caches generally have greater latency and consume more chip area.

2.Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.

SRAM and DRAM are both types of volatile semiconductor memory, but they are designed for different purposes.

Comparison

Feature	**SRAM**	**DRAM**
Full form	Static RAM	Dynamic RAM
Storage element	Flip-flop	Capacitor + transistor
Latency	Very low	Higher
Bandwidth	Very high	High

Throughput	Very high	High
Refresh	Not required	Required
Density	Low	High
Cost	Expensive	Less expensive
Power	Higher area-related cost	Requires refresh power
Typical use	CPU cache	Main memory

SRAM
SRAM stores each bit using a circuit based on flip-flops. Once data is stored, it remains available as long as power is supplied.
It does **not require periodic refreshing**.

Advantages:
- Very fast access
- Low latency
- High throughput
- Suitable for frequent CPU accesses

Disadvantage:
- Requires more transistors per bit
- Takes more chip area
- More expensive

Therefore, SRAM is used for:
L1 Cache → L2 Cache → L3 Cache

DRAM
DRAM stores each bit as an electrical charge in a capacitor. The charge gradually leaks away, so DRAM must be periodically refreshed.

Advantages:
- High storage density
- Lower cost per bit
- Large capacities are possible

Disadvantages:
- Higher latency
- Requires refresh operations
- Generally slower than SRAM
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DRAM is therefore commonly used as **main memory**.

Why SRAM for Cache?
The CPU requires extremely fast access to frequently used data. SRAM provides the required low latency and high throughput.

Why DRAM for Main Memory?
The system requires much more main memory than cache. Using SRAM for several gigabytes of RAM would be very expensive and require a large amount of chip area.
Therefore:
SRAM → Small + Fast + Expensive
DRAM → Large + Slower + Cheaper

Conclusion
SRAM is best for **cache memory**, while DRAM is best for **main memory**. Their combination provides an effective balance between **performance, capacity, and cost**.

3.Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.

Cache memory and virtual memory both improve memory management, but they solve different problems.

Basic Concept

Cache:
CPU → Cache → Main Memory
Virtual Memory:
CPU → Virtual Address → Page Table → Physical Memory → Secondary Storage

Virtual memory allows a system to use secondary storage as an extension of physical memory.

Comparison

Feature	Cache Memory	Virtual Memory/Paging
Main purpose	Reduce memory access latency	Provide larger logical memory
Managed by	Hardware/cache controller	OS + MMU
Storage	SRAM	DRAM + SSD/HDD
Unit	Cache line/block	Page
Latency	Very low	Higher
Page fault	No	Yes
Speed	Extremely fast	Much slower when storag involved
Effect on throughput	Generally increases	Can decrease during excessive p

Cache Memory
Cache stores frequently accessed data close to the CPU.
For example, if a loop repeatedly accesses the same array elements, those elements can remain in cache.
This reduces the time required for memory access.

Virtual Memory
Virtual memory allows a program to use an address space larger than the available physical RAM.
Memory is divided into fixed-size blocks called pages.
When a required page is not present in RAM, a page fault occurs. The operating system must bring that page from secondary storage into RAM.

Why Page Faults are Expensive
The speed difference between RAM and storage can be very large.
Therefore:
Cache hit → Very fast
RAM access → Slower
Page fault → Extremely slow compared with RAM
If page faults occur frequently, the system can enter thrashing, where the operating system spends excessive time moving pages rather than executing programs.
Effect on System Throughput
Cache memory usually increases throughput because the CPU spends less time waiting for memory.
Virtual memory increases the effective memory capacity, but excessive paging can reduce throughput.

Conclusion
Cache memory is primarily a performance optimization, whereas virtual memory is primarily a capacity and memory-management mechanism.

4.Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.

NAND Flash and DRAM are both semiconductor memory technologies, but they occupy different levels of the memory hierarchy.

Comparison

Feature	**NAND Flash**	**DRAM**
Volatility	Non-volatile	Volatile
Latency	Higher	Much lower
Read speed	High	Very high
Write speed	Relatively slower	Faster
Bandwidth	High, especially in SSDs	Very high
Throughput	High for storage workloads	Very high for memory workloa
Refresh	Not required	Required
Capacity	Very high	High
Endurance	Limited write/erase cycles	Very high

Typical use	SSDs, storage	Main memory

DRAM
DRAM is used for active program data and instructions.
The CPU frequently accesses DRAM when the requested information is not available in cache.
DRAM provides:

- Low latency
- High bandwidth
- High random-access performance

Therefore, it is suitable for **main memory**.

NAND Flash
NAND Flash is non-volatile, meaning it retains information even after power is removed.
It is commonly used in:

- SSDs
- USB drives
- Memory cards
- Embedded storage

NAND Flash offers high storage density and relatively low cost per bit.
However, it has higher access latency than DRAM, particularly for writes and random accesses.

Suitability
DRAM is suitable when **speed and frequent random access** are required.
NAND Flash is suitable when **large capacity and data persistence** are more important than extremely low latency.

Conclusion
DRAM is better for **active working memory**, while NAND Flash is better for **persistent storage**. Both technologies complement each other in a hierarchical memory system.

5.Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

MRAM and RRAM are emerging/non-volatile memory technologies that can potentially reduce the gap between conventional memory and storage.

MRAM = Magnetoresistive Random Access Memory
MRAM stores information using magnetic states rather than electrical charge.
Because it is non-volatile, data remains stored even when power is removed.

Advantages of MRAM

- Fast access
- Non-volatile
- High endurance
- Good energy efficiency
- Suitable for embedded applications

MRAM can potentially be used in:

- Embedded memory
- Microcontrollers
- Cache-like applications
- Low-power systems

RRAM = Resistive Random Access Memory

RRAM stores information by changing the resistance of a material.
Different resistance states represent different data values.

Advantages of RRAM

- Fast switching potential
- Non-volatile
- High density potential
- Low standby power
- Potentially good energy efficiency

RRAM is being investigated for:

- Embedded memory
- High-density memory
- Storage-class memory
- Neuromorphic computing

Comparison

Feature	**MRAM**	**RRAM**
Storage principle	Magnetic state	Resistance state
Volatile?	No	No
Access time	Very fast	Very fast potential
Read latency	Low	Low
Throughput	High	High potential
Energy efficiency	High	Potentially very high
Endurance	Very high	Technology-dependent
Density	High	Potentially very high
Data retention	Good	Good
Main advantage	Speed + endurance	Density + low-power potential

Access Time
MRAM can provide access times closer to conventional fast memory than many traditional non-volatile memories.
RRAM also has the potential for very fast switching because it changes between resistance states.

Throughput
Both technologies can provide high throughput. Actual performance depends heavily on the particular device architecture, memory array, controller, and fabrication technology.

Energy Efficiency
MRAM does not require continuous refresh, unlike DRAM. This can reduce standby energy.
RRAM can also have very low standby energy because it is non-volatile. Its high-density structure can further improve energy efficiency per stored bit.

Conclusion
MRAM is attractive when **high speed, endurance, and reliability** are important. RRAM is attractive when **high density, low standby power, and future scalable memory architectures** are priorities.

Code of Conduct:

I S.Gayatheri(192521306) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
S.GAYATHERI

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand